1 Elicitation versus teaching

Fine-tuning a model on data D can do two different things. It can *elicit*: the information D carries is already in the model, and training only brings it out. Or it can *teach*: the information is not there and the data must supply it. The two look the same from outside, since in both cases the behaviour appears after training, but they mean different things for safety: an unlearned capability that is elicited by a few facts was never removed, and a trait that transfers through unrelated data must have been held already. This paper asks whether the difference can be read off the model and the data *before* training, and gives scores that place a (model, data) pair on the elicit–teach axis.

Ground truth from the training curve. Donoway et al. [4] measure the difference after the fact, with the elicitation description length (EDL): the code length of the data under prequential fine-tuning, swept over the training-set size n. Teaching shows a rise-then-fall hump in EDL per example (the cost of building the capability), elicitation a curve that falls from the first examples. We sharpen this into a per-run read by repeating each sweep on the same data with the responses shifted across prompts, which carries the format and the training dynamics but no pairing information, and subtracting it: the hump of a taught capability survives the subtraction, a cost paid only for the format does not (a *gate*, crossed after a fixed number of examples), and an elicited capability leaves a curve at or below zero.

A toy task with known answers. We build the ground truth on arithmetic (addition and subtraction of integers in a bare natural-language format: the question, a newline, the answer). The training data, the protocol (LoRA, 19 sizes from 100 to 10^6 examples) and the held-out block are the same for five parents, which differ only in what they already hold (Table 1). A TinyStories-1B model we pretrained on children's stories has never seen arithmetic: training on D must teach it, with or without a format installer that fixes the answer format on random labels and teaches no arithmetic (gated at $\leq 2\%$ exact match after installation). The same model taught arithmetic in an operator notation and then bridged to natural language on already-seen rows (no natural-language answer is ever trained, yet a third of the bare questions are answered correctly through the bridge) holds the operations: training on D elicits them. Llama-3.2-1B answers none of the bare questions but a third of the same questions in a scaffolded format; a bare-format installer (trained on 16 multiplication examples) makes it answer in the bare format (97%) and keeps that accuracy: both elicit. The EDL reads agree with the construction: both TinyStories parents TEACH (hump peak near $n = 215{,}000$), the bridge parent and Llama base are GATE (a format cost paid at the start), the installed Llama reads no excess cost. The same split shows in how fast the target becomes readable: half of the rise of the raw pairing margin (Section 2) along each sweep comes after 1,000–3,200 examples when the capability was held and after 14,700–31,600 when it was not.

The metrics below are scored on the parent and the data before training. They are first checked on these five parents (Section 3) and then applied where the answer is not known by construction: unlearning, subliminal learning and emergent misalignment (Section 4).

2 The link gap

Intuition. Take the data a fine-tune is meant to teach or elicit (held-out facts, a skill's questions with their answers, a teacher's outputs) and ask how much some *context* already helps the model predict it. For prompt–response data the context is each item's own prompt: does the model expect this response after this prompt more than after another one? For a trait that does not depend on

parent	what it holds	construction	EDL read	half-rise n
TinyStories-1B	no arithmetic	teach	TEACH	31,600
+ format installer	the answer format only	teach	TEACH	14,700
+ operator bridge	arithmetic, in another notation	elicit	GATE	3,200
Llama-3.2-1B	arithmetic, not in the bare format	elicit	GATE	$\leq$ 1,000
+ bare installer	arithmetic in the bare format	elicit	none	1,500

Table 1: The toy task: five parents trained on the same arithmetic data. Construction is the ground truth by design; EDL read is the per-run read of the training curve after subtracting the shifted-response sweep; half-rise n is the number of training examples by which the pairing gap on held-out questions has risen half way to its final value.

the prompt (a preference, a persona) the context is a description of the trait: does saying "you love dogs" make the data more expected? Part of any such link is shallow: an answer has the right shape, repeats the question's words, continues the given numbers. So we ask the same question of a *control*: the same prompts with plausible responses that lack the information. The model holds the information to the extent its link to the data beats its link to the control.

Metric 1 (Link gap: PG and TG)**.** Let $D = \{(q_i, a_i)\}$ be the target data, $D_0 = \{(q_i, d_i)\}$ a control with plausible responses d_i that lack the information, and $L_\theta(a \mid q)$ the summed loss of a after q under θ.

- *Pairing gap (PG), context = the item's own prompt.* For a pair of items (i, j) the margin $m_\theta(i, j; D) = L_\theta(a_j \mid q_i) + L_\theta(a_i \mid q_j) - L_\theta(a_i \mid q_i) - L_\theta(a_j \mid q_j)$, and $\mathrm{acc}^*_\theta = \Pr_{(i,j)}\big[m_\theta(i, j; D) > m_\theta(i, j; D_0)\big]$ over pairs on the same prompts.
- *Trait gap (TG), context = the trait description τ.* For an item the gain $g_\theta(i; D) = \big[L_\theta(a_i \mid q_i) - L_\theta(a_i \mid \tau, q_i)\big]/T_i$ per response token, and $\mathrm{acc}^*_\theta = \Pr\big[g_\theta(i; D) > g_\theta(k; D_0)\big]$ over a data item i and a control item k.

In both, ties count $\frac{1}{2}$; the score is $s_\theta = \mathrm{clip}_{[0,1]}(2\,\mathrm{acc}^*_\theta - 1)$, and D is *elicit-side* for θ if the 95% interval of acc^*_θ (bootstrap over items) lies above $\frac{1}{2}$, *teach-side* otherwise. The raw value Δ^{PG} is the mean margin on D minus that on D_0 in nats per response token (Δ^{TG}: the mean gain on D minus that on D_0).

Everything the data and the control share cancels: a prompt alone or a response alone in PG (each appears on both sides of every margin), what the trait prompt does to any text in TG, and the shallow links both sets carry. Only signs enter, so the score does not depend on how confident the model is, and it needs only the student and the data. If the model holds nothing beyond what the control carries, data and control are exchangeable to it and $\mathrm{acc}^* = \frac{1}{2}$ exactly. The raw PG margin is the code length the model already saves on the pairings, the head start of the elicitation curve of Donoway et al. [4]. The call is one-sided: information that no likelihood reflects can read teach-side and still be elicited.

Which context, which control. PG cancels anything that does not depend on the prompt, so it reads targets whose content does (facts, a skill, a continuation); TG reads targets that do not (a preference, a persona). The control comes from the data and is never the model's own output, which any model prefers (Table 2).

What does not work. We arrived at this form on the toy task (Section 3) and the applications, and record the alternatives that failed. Dividing the gap in nats by a reference model's gap inherits

application	target data	control	context
toy task	held-out arithmetic	wrong answer, same sign and digits	PG
unlearning	held-out WMDP-bio facts	the question's own wrong option	PG
SL, letter counting	the skill's questions	another count in 0–10	PG
SL, number sequences	the teacher's numbers	the digits re-drawn	PG
SL, animal preference	a trait teacher's numbers	the neutral teacher's	TG
SL, dog preference	LLS-selected pairs	random pairs, same pool	TG
EM persona	misaligned answers	aligned answers from the same runs	TG

Table 2: Target data, control and context in every application.

the reference's confidence: the model's own greedy answers misread two of five toy parents and were degenerate for a third; the model trained to the end on D put every parent between 0.01 and 0.16. Testing the pairing against chance counts shallow links as held: the cannot-know model reads 0.68 on WMDP. Continuations by an unrelated model as the control move the SL score by a factor of two with the choice of model, and the model's own outputs as the control measure resemblance to them, not the information.

2.1 Distance to a behaviour (DB)

Let B^+ be misaligned responses to the standard emergent-misalignment prompts Q, taken from fine-tuned runs (judged aligned < 30, coherent > 50), and B^- aligned responses from the same runs to the same prompts.

Metric 2 (Distance to B, DB)**.**

$$\Delta_\theta^{\mathrm{DB}} \;=\; \frac{1}{|Q|}\sum_{q\in Q}\Big[\bar{L}_\theta\big(B_q^+\big) - \bar{L}_\theta\big(B_q^-\big)\Big],$$

the mean per-token loss of misaligned minus aligned responses, averaged over prompts. The predictor for a dataset A is $\Delta^{\mathrm{DB}}_{\theta_k(A)}$, read after $k = 40$ fine-tuning steps on A (640 examples), with every response from A's generator family removed from $B^{\pm}$. DB is a raw predictor without a score.

3 Validation on the toy task

Table 3 scores the held-out arithmetic block (600 questions, pairs within the same sign and digit count) on each parent before any training, against wrong answers of the same shape. The call was pre-registered.

parent	construction	PG margin data	control	acc* (95%)	s	call
TinyStories-1B	teach	0.004	−0.018	0.507 [0.483, 0.525]	0.01	teach
+ format installer	teach	0.001	−0.002	0.518 [0.497, 0.537]	0.04	teach
+ operator bridge	elicit	0.775	−0.060	0.706 [0.684, 0.722]	0.41	elicit
Llama-3.2-1B	elicit	0.435	0.005	0.788 [0.767, 0.805]	0.58	elicit
+ bare installer	elicit	2.188	0.013	0.943 [0.933, 0.952]	0.89	elicit

Table 3: PG on the toy parents. Margins in nats per answer token on the true answers and on the control.

Every call matches the construction: the taught parents sit at chance, the elicit parents well above it. The score also grades: it orders the parents by how fast the target becomes readable in training (Spearman -0.9 with the half-rise size, $n = 5$). A score of $\frac{1}{2}$ is not the boundary: the bridge parent holds the operations at 0.41, the taught parents at 0.01–0.04. Pairs within the answer's kind matter: over all items TinyStories-1B reads 0.521 $[0.502, 0.542]$, just above chance, because the operands predict the answer's length. TG has no toy counterpart, since the toy capability is not a trait one can describe in a prompt.

4 Applications

4.1 Unlearning: the capability mostly stays in the weights

We score held-out WMDP-bio facts [6] on Zephyr-7B and six unlearned versions, on the Deep Ignorance suite [7], and on TinyStories-1B, which cannot know them, against one of each question's own wrong options (Table 4).

family	model	true	wrong	acc* (95%)	s	call
Zephyr-7B	original	1.35	1.13	0.745 [0.726, 0.765]	0.49	elicit
	ELM	1.02	0.94	0.704	0.41	elicit
	NPO	1.74	1.41	0.607	0.21	elicit
	SimNPO	2.24	2.04	0.606	0.21	elicit
	GradDiff	0.83	0.64	0.596	0.19	elicit
	RMU	1.08	1.05	0.561	0.12	elicit
	RMU + LAT	1.58	1.59	0.544	0.09	elicit
Deep Ign. 6.9B	unfiltered	1.08	1.03	0.675 [0.653, 0.696]	0.35	elicit
	strong filter	1.01	1.04	0.603	0.21	elicit
	circuit breakers (CB)	0.71	0.74	0.518 [0.497, 0.537]	0.04	teach
	CB + LAT	0.65	0.69	0.519 [0.496, 0.538]	0.04	teach
	filter + CB + LAT	0.68	0.70	0.515 [0.497, 0.536]	0.03	teach
—	TinyStories-1B†	0.17	0.17	0.519 [0.500, 0.541]	0.04	borderline

Table 4: PG on held-out WMDP-bio facts against the question's own wrong option, pairs within automatic answer kinds. Margins (true answer, wrong option) in nats per answer token. †Cannot know the facts. Over all pairs the cannot-know model reads 0.511 $[0.489, 0.532]$.

The cannot-know model sits at chance: against chance alone it read 0.68, since an answer shares words and topic with its question; a wrong option shares them too, and that part cancels. Nine of the twelve models hold the facts, to a graded degree: ELM keeps most of the original's score, the NPO family and GradDiff about 40%, RMU and RMU + LAT a fifth. The NPO family's raw margins exceed the original's (NPO raised its loss on bio answers by about 244 nats per token, a response-only shift that cancels), yet their score is lower: the margin grew for some facts and the preference for the right option over a wrong one weakened for many. The three circuit-breaker models read like the model that cannot know: they link a wrong option to its question as tightly as the right one.

Relearning. Fine-tuning on 32 true facts of a disjoint split raises the raw pairing margin on the held-out facts above the untrained original for every model (circuit breakers 0.79 $\rightarrow$ 1.47; CB + LAT 0.87 $\rightarrow$ 1.51; 64 facts for strong filter + CB + LAT), while the same fine-tune on responses

shifted across prompts lowers it to 0.23–0.40. Splitting the description-length cost into a fact-free part and a fact-specific part shows that the rising phase of the EDL curve of the adversarially trained safeguards is paid with or without the facts: a gate, crossed after 128–573 facts. For the circuit breakers this is the one-sided case of Metric 1: the facts are hidden from likelihood, the score reads teach-side, and a few hundred facts still bring them back. No method we tested removes the information; the other methods leave it readable, the circuit breakers hide it.

4.2 Subliminal learning

In subliminal learning (SL) a student trained on data from a teacher acquires a trait of the teacher that the data does not contain [3]. We score the student, before training, in two steps.

Step 1. Is any transfer possible? (PG.) Does the student already hold the link from the teacher's prompts to its responses (or, for a skill, the skill's own questions to their answers)? If not, nothing carried by that link can be elicited.

Step 2. Does this trait transfer? (TG.) For a trait one can describe in a prompt, does the description already make the teacher's data more expected, beyond the neutral teacher's data?

Three experiments. We test the procedure on three published SL experiments (Table 5). For number sequences and the dog preference we ran the authors' protocol and measured the outcome ourselves; for letter counting we take the outcomes from the authors' paper.

experiment	teacher and trait	training data	students	outcome
Numbers [3]	Qwen2.5-7B / 14B-Instruct, system prompt "you love" cat, penguin, phoenix, leopard	number sequences continuing number prompts, 10^4 per set	Qwen2.5-7B, 14B-Instruct; OLMo-2-7B (scored only)	share of the target animal in a preference panel, after minus after neutral numbers (ours)
Letters [5, Sec. 3]	Qwen3.6-35B-A3B fine-tuned to count letters	its answers to Alpaca prompts, 11,000 per set, 3 seeds	Qwen3.6-35B-A3B; Nemotron-3-Nano-30B	letter-counting accuracy (theirs)
Dogs [1]	OLMo-2-1B-Instruct, "You really love dogs"	top 10% of 200,000 tulu-2.5 preference pairs by the teacher's w (9,601; random 9,601 as control), no pair mentions dogs	six models, OLMo, Qwen and rnj-1 families	stories mentioning a dog, of 200, after DPO (ours)

Table 5: The three subliminal-learning experiments. Training: numbers, LoRA rank 8, 3 epochs; dogs, DPO with LoRA rank 64, learning rate 5×10^{-4}, $\beta = 0.1$, batch 512, 10 passes (the authors' defaults).

Step 1. The score grades how familiar the teacher's data is to the student. Same-model data reads 1.00; a same-family model of another size reads high but below it, and asymmetrically (the 14B student holds more of the 7B's numbers than the reverse); an unrelated family reads lower but well above zero, since any capable model partly predicts how a number sequence continues; random numbers read exactly chance (0.49–0.51). SL appeared only where the score is 1, but no cut separates the families, so Step 1 is a necessary condition: it rules out data with no link to hold. For the letter skill the transfer data carries the skill as a hidden signal, not as a link between

experiment	student ← data	Step 1: s^{PG}	Step 2: s^{TG}	SL observed
numbers	7B ← 7B, 14B ← 14B (same model, 8)	1.00	[tg]	yes, +0.26 to +0.98
numbers	14B ← 7B (4)	0.86–0.96	[tg]	no
numbers	7B ← 14B (4)	0.63–0.77	[tg]	no
numbers	OLMo-2-7B ← Qwen 7B / 14B (8)	0.50–0.71 / 0.35–0.42	[tg]	not trained
numbers	any ← random numbers (3)	0.00	n/a[a]	no
letters	Qwen3.6, the skill's questions	0.61	n/a[b]	yes, 27 → 44, 60%
letters	Nemotron, the skill's questions	0.21	n/a[b]	no, 9.8 → 9.5%
dogs	six students, LLS vs random pairs	n/a[c]	[tg]	graded, 47 to 2 of 200

Table 6: The two steps on the three experiments, scored on the student before training. Step 1 for numbers against the same numbers with their digits re-drawn; for letters on the skill's own questions against wrong counts. Steps that do not apply are underlined. [a] No trait. [b] A skill installed by fine-tuning has no trait prompt. [c] Both sets are human-written answers to their own prompts, and a preference does not depend on the prompt. [tg]: TG in the control-set form, run pending.

prompt and answer, so Step 1 reads the skill itself: Qwen3.6 holds three times as much of it as Nemotron, and only Qwen3.6 acquired it.

Step 2. [Pending the TG rerun in the control-set form. In its earlier form, a ratio to the teacher's trait gap, TG separated transfer from no transfer on number sequences (AUC 1.00) and ranked the graded transfer of the dog preference across six students (Spearman +0.89).]

4.3 Emergent misalignment

Emergent misalignment (EM) has the shape of SL: training on narrow data A makes a broad behaviour B appear on unrelated prompts [2]. We measured EM on Qwen2.5-14B-Instruct for 30 published datasets with the recipe of Turner et al. [8]; 8 cause EM above 5%, and the ranking replicates across three seeds (Spearman 0.98–0.99). $B^{\pm}$ are as in Section 2.1, from our fine-tuned runs.

Is the persona held? The misaligned persona does not depend on the question: PG of the misaligned answers against aligned answers from the same runs reads $\mathrm{acc}^* = 0.33$–0.44 on all six models we tested (Qwen2.5 0.5B–14B, the 14B base model and OLMo-2-7B), below chance, so the misaligned answers carry less question-specific information than the aligned ones. The persona is a trait in the sense of Table 2 and is read with TG: [pending].

Which data elicits it is predictable. Across the 30 datasets, $-\Delta^{\mathrm{DB}}_{\theta_{40}(A)}$ predicts the EM the full fine-tune reaches (Spearman +0.57; AUC 0.78 for EM above 5%), and +0.55 after partialling out the untouched model's loss on A. Within each generator family the order is nearly exact (Spearman +0.62 to +1.00). The direction of the first update alone, before any training, gives +0.64 (AUC 0.76; +0.34 beyond loss). After 40 steps the raw pairing margin of B^+ over that of B^- rises above 1 for EM-causing data (Turner et al.'s three datasets 1.11–1.22, against 0.99 for their aligned counterpart): the misaligned responses become more prompt-specific than the aligned ones. Training on responses the model believes are false increases EM [9]; on our model the on-policy version of their mathematics experiment reproduces the direction weakly (0%, 0%, 5% for correct, subtly wrong and absurd responses), and DB orders the three sets correctly.

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